

Saliency-LOAM: Saliency-Based LiDAR Odometry and Mapping

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Abstract—Simultaneous localization and mapping (SLAM) is a critical research topic in autonomous driving, robotics, and related fields. Although LiDAR-based SLAM methods achieve excellent performance, they primarily extract feature points from the geometric structure of point clouds, failing to mitigate the adverse effects of noise interference on feature matching, which leads to inaccurate localization in complex environments. In this article, we present Saliency-LOAM, a novel LiDAR odometry method based on point cloud saliency. The framework proposes Saliency-Net, a feature extraction network that integrates geometric, intensity, and semantic information of point clouds to effectively predict the saliency information of different regions in 3-D scenes. It assigns higher saliency weights to regions with greater feature stability. During the feature matching process, saliency weights are utilized to adjust the distance residual term, ensuring that the optimization equation focuses on more stable and higher-quality point cloud regions, thereby iteratively optimizing to achieve more accurate pose estimation. Experimental results on public datasets and real-world scenarios show that, compared to existing methods, the point cloud saliency mechanism better suppresses the negative impact of noise in raw point cloud data on point cloud matching, significantly improving the robustness and accuracy of LiDAR SLAM systems. Our code will be open-sourced at <https://github.com/ck-png/Saliency-LOAM>

Index Terms—LiDAR odometry, point cloud saliency, simultaneous localization and mapping (SLAM).

I. INTRODUCTION

IN THE field of autonomous driving, precise environmental perception and reliable real-time localization are crucial for achieving advanced autonomous driving systems. Currently, simultaneous localization and mapping (SLAM) has become an indispensable key technology due to its unique advantages. Given the differences in sensor types,

LiDAR-based SLAM offers significant advantages over visual SLAM [1], [2]. Regardless of lighting or weather conditions, LiDAR sensors provide consistent and accurate data. Existing LiDAR SLAM methods [3], [4], [5], [6] primarily rely on geometric features of point clouds to estimate pose and construct maps. Researchers explore additional point cloud information, including intensity and semantic information, to assist in extracting more stable features and enhance the performance of SLAM systems.

Saliency, initially popularized in visual tasks, is a mechanism that emulates human visual attention, whereby humans focus on prominent objects and automatically assign higher weights to such targets when performing self-localization through visual perception. For instance, when navigating unknown environments, humans tend to focus on distinctive landmarks such as buildings or signs to determine their position, as these objects exhibit prominent and stable features. In the field of visual SLAM, our prior work SBAS [7] computes saliency maps by incorporating geometric and semantic information from images, ultimately enhancing the performance of visual odometry. In LiDAR-based SLAM, studies [8], [9] integrate geometric and intensity information of point clouds and design distinct weighting functions for line and plane feature points, effectively suppressing uncertainties in feature matching introduced by noisy point clouds. However, relying solely on geometric and intensity information limits a comprehensive understanding of the environment. Semantic information provides richer feature representations, offering the potential to improve localization accuracy and enable more intelligent functionalities for downstream planning tasks [10]. Therefore, we propose Saliency-LOAM, which, in addition to geometric information, further integrates intensity and semantic information of point clouds. The detailed analysis is as follows.

Geometric information enables the initial extraction of line and plane features in the environment, although it still includes noise. Intensity information reflects the material properties and light reflectance capabilities of different objects, with distinct objects exhibiting strong gradient changes at edges. These changes are stable, facilitating the identification of more reliable line and plane features and filtering out extraneous noise. Furthermore, the semantic information of point clouds provides high-level environmental understanding, allowing focus on objects such as vehicles, trees, and utility poles, which possess more stable geometric structures compared to other point clouds. This addresses the limitation of traditional

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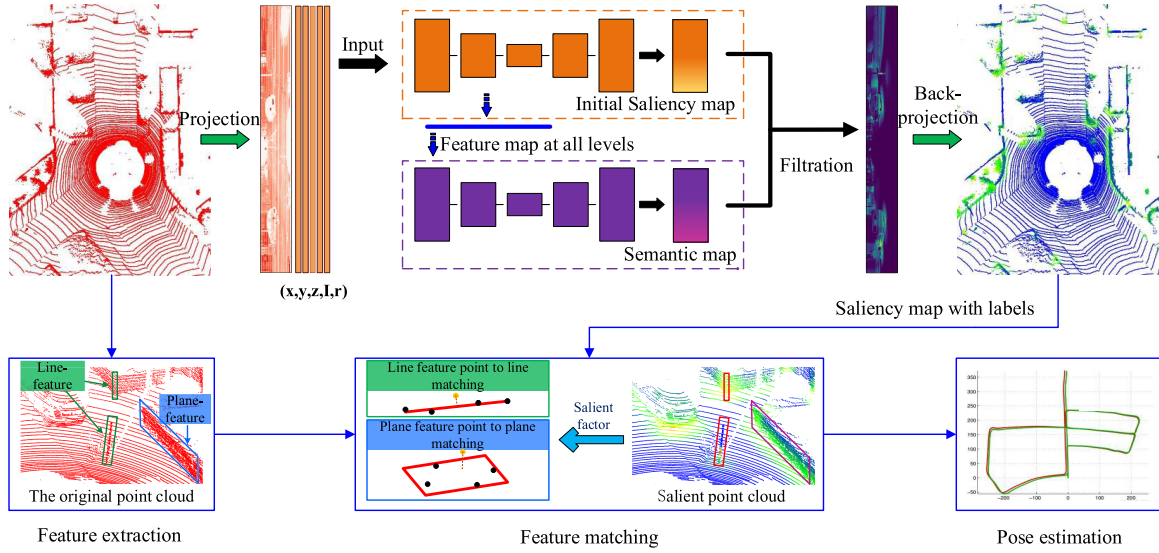


Fig. 1. Saliency-net module (the part with orange and purple) takes the projected 2-D images as input and generates the saliency point cloud through back-projection. The feature extraction module extracts line features and plane features. Based on the saliency values provided by the saliency point cloud, the saliency feature matching module minimizes the distance between line feature points and lines, as well as the distance between plane feature points and planes through iterative optimization, which emphasizes the optimization of more salient and critical features. Finally, the pose estimation module performs pose estimation based on the matching results.

line and plane features, which fail to leverage advanced geometric structures in the environment. To this end, we propose Saliency-Net, which effectively integrates these three types of information to compute saliency information. This approach not only extracts more stable line and plane features but also assigns higher weights to them during feature matching, thereby enhancing the accuracy and robustness of LiDAR SLAM systems.

This article proposes Saliency-LOAM, a LiDAR-based SLAM method that introduces the concept of saliency to the field of LiDAR odometry for the first time, enabling accurate, robust, and real-time localization and mapping. Specifically, our contributions are as follows.

- 1) We propose Saliency-Net, a neural network designed to predict point cloud saliency, integrating geometric information, intensity information, and semantic information from LiDAR point clouds. To better guide network training, we create a dataset with saliency ground truth based on these information types. Through supervised training, Saliency-Net effectively predicts point cloud saliency in autonomous driving scenarios.
- 2) Based on Saliency-Net, we propose Saliency-LOAM, a LiDAR SLAM method incorporating the point cloud saliency mechanism. In the feature matching stage, the algorithm utilizes predicted saliency weights to constrain distance residual calculations, enabling it to focus on feature-stable point cloud regions, thereby enhancing the robustness and accuracy of point cloud feature matching and pose estimation.
- 3) Experimental results on open-source datasets demonstrate the effectiveness of the point cloud saliency mechanism, with our method outperforming state-of-the-art approaches in pose estimation performance and mapping quality. To validate practical applicability, we

conduct real-vehicle comparison tests based on a campus environment. Our method achieves superior performance across numerous challenging scenarios.

II. METHOD

As shown in Fig. 1, the proposed Saliency-LOAM consists of four main modules: feature extraction, point cloud saliency prediction, saliency feature matching, and pose estimation.

A. Design of Saliency Weights

Geometric information includes distance information and local curvature. For a point $P_i(x_i, y_i, z_i, I_i)$, where x_i , y_i , and z_i are the 3-D coordinates and I_i is the intensity value, distance information is obtained using the following formula, which reflects the spatial structure and geometric shape of the 3-D point cloud, facilitating the acquisition of local curvature. This also ensures alignment between saliency information and the original point cloud, while supporting the extraction of semantic information

$$r_i = \sqrt{x_i^2 + y_i^2 + z_i^2}. \quad (1)$$

For local curvature, we utilize a local smoothness measure, denoted as c_i , to represent the local curvature. Taking n points adjacent to the current point i to form a point set N , setting r_j as the distance information of the j th point, then the local smoothness c_i of the current point i is calculated as follows:

$$c_i = \frac{1}{2n \cdot \|r_i\|_2^2} \left\| \sum_{j \in N, j \neq i} (r_j - r_i) \right\|_2^2. \quad (2)$$

Specifically, during the filtering process, thresholds TH_{line} and TH_{plane} need to be set for line and plane features, respectively. Points with local curvature $c_i \geq TH_{line}$ are selected

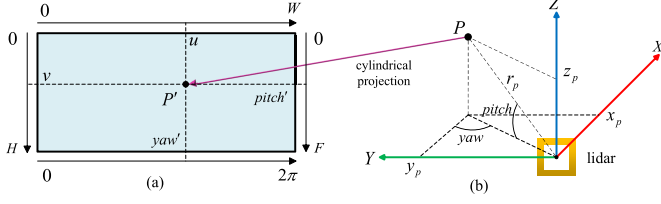


Fig. 2. Cylindrical projection of point cloud. (a) Point cloud cylindrical graph. (b) Point cloud 3-D coordinate system.

as line feature points, while those with $c_i \leq \text{TH}_{\text{plane}}$ are considered plane feature points.

For intensity information, we use K-nearest neighbors (KNN) to identify the neighborhood $\mathcal{N}_i$ of point P_i . Taking the x -direction as an example, the intensity gradient is computed as follows:

$$\frac{\partial I}{\partial x} = \sum_{p_j \in \mathcal{N}_i} w_{ij} \cdot (I_i - I_j) \cdot \frac{x_j - x_i}{\|p_j - p_i\|} \quad (3)$$

where w_{ij} is the Gaussian weight, accounting for the contribution of spatial distance to the gradient, defined as

$$w_{ij} = \exp\left(-\frac{\|p_j - p_i\|^2}{2\sigma^2}\right) \quad (4)$$

where p_j is a neighboring point, I_j is the intensity value of point p_j , and x_j is the coordinate of p_j . The above computation yields the gradient vector as follows:

$$\nabla I_i = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}, \frac{\partial I}{\partial z} \right). \quad (5)$$

Subsequently, the magnitude of the intensity gradient is obtained as

$$\|\nabla I_i\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2 + \left(\frac{\partial I}{\partial z}\right)^2}. \quad (6)$$

By setting an appropriate threshold, more stable point cloud regions are selected for subsequent processing.

The Semantic KITTI dataset [11] provides comprehensive semantic labels. Since point clouds belonging to semantic categories exhibit greater feature stability than nonsemantic point clouds, while dynamically semantic point clouds tend to suppress feature matching, we assign the semantic weight M_{sem} a uniform weight of 1 to statically semantic point clouds, dynamically semantic point clouds and nonsemantic point clouds are assigned a weight of 0.

As shown in Fig. 2, the number of lasers is the image height H , and the top and bottom pitch angles are F_{Up} and F_{Down} , the field of view of LiDAR can be obtained as $F = F_{\text{Up}} - F_{\text{Down}}$, and the width of the image W is the number of times the laser scans a week. We obtain the distance map, local curvature map, semantic map, and intensity map of the point cloud according to the following formula:

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \left[1 + \arctan\left(\frac{y_p}{x_p}\right) \pi^{-1} \right] W \\ 1 - \left[\frac{z_p}{\sqrt{x_p^2 + y_p^2 + z_p^2}} - F_{\text{Down}} \right] F^{-1} \end{bmatrix} H. \quad (7)$$

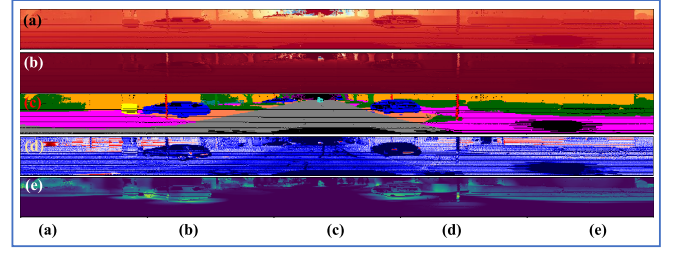


Fig. 3. Projection maps of various point cloud information. (a) Distance map. (b) Local curvature map. (c) Semantic map. (d) Intensity map. (e) Saliency map.

Then, image pyramids are computed separately for distance and intensity images. Gaussian filtering is then applied to the images of different sizes. Subsequently, ITTI [12] saliency computation is performed on both the filtered and prefiltered images to obtain distance feature maps D_s and intensity feature maps I_s for the point cloud.

The projected data points within a cell (u, v) are defined as a set. For the intensity feature map I_s , the average intensity gradient variation $I_{\text{avg}}(u, v)$ within each cell is computed as the confidence weight. For the distance map D_s , the local distance variance $\text{Var}_r(u, v)$ within each cell is calculated, as dense regions often exhibit distinctive features. An exponentially decaying distance weighting scheme is also incorporated to impose a constraint, since points closer to the sensor are generally more important; the two factors are combined to form the confidence measure. For the local curvature, the average curvature within each cell is directly computed as $C_{\text{avg}}(u, v)$. The three types of information are individually normalized and then weighted, and finally summed with the semantic weight $M_{\text{sem}}(u, v)$, which is 1 or 0. Thus, the saliency weight $S(u, v)$ for cell (u, v) is given by

$$S(u, v) = \text{Norm}(\text{Norm}(I_{\text{avg}}(u, v) \cdot I_s(u, v)) + \text{Norm}(w_d(u, v) \cdot D_s(u, v)) + \text{Norm}(C_{\text{avg}}(u, v)) + M_{\text{sem}}(u, v)) \quad (8)$$

$$w_d(u, v) = \exp\left(-\frac{\text{Var}_r(u, v)}{\sigma_v^2} - \frac{\mu_r(u, v)^2}{2\sigma_d^2}\right) \quad (9)$$

where $\text{Var}_r(u, v)$ is the local distance variance, $\mu_r(u, v)$ is the mean distance, and σ_v^2 , σ_d^2 are scaling parameters.

Norm refers to the standard Min-Max normalization method, which linearly maps different features to a unified interval $[0, 1]$. For a given input set, where x is the current value, $\max(x)$ is the maximum value, and $\min(x)$ is the minimum value, the normalization is defined as follows:

$$\text{Norm}(x) = \frac{x - \min(x)}{\max(x) - \min(x)}. \quad (10)$$

Since feature data of different types exhibit varying orders of magnitude, single-layer normalization may significantly weaken features with smaller scales. In contrast, our two-layer normalization approach eliminates biases caused by differences in physical dimensions and numerical ranges. In summary, we utilize multiple types of point cloud information to identify salient point clouds, followed by weighted fusion to

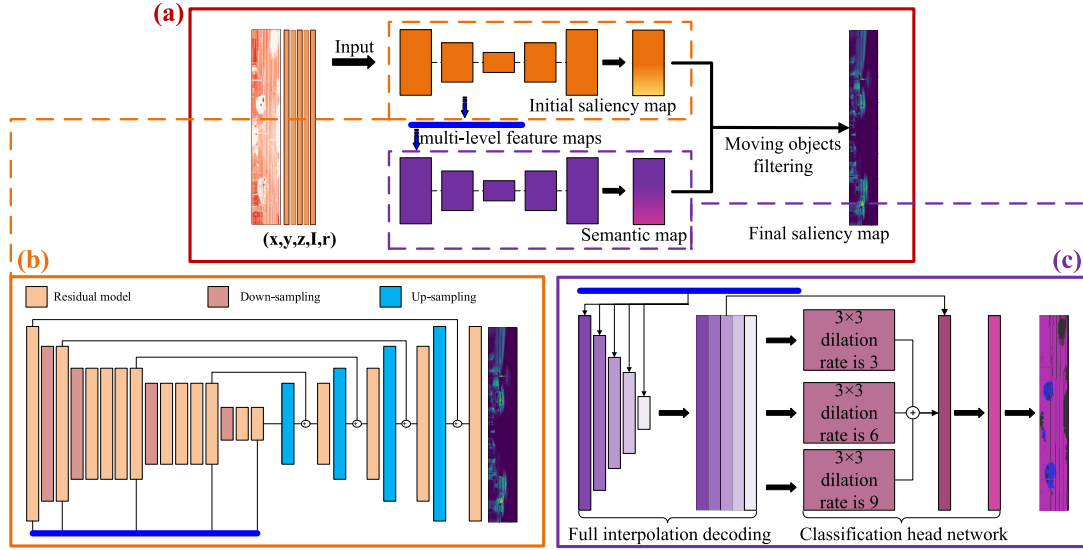


Fig. 4. Saliency-net: point cloud saliency prediction framework. (a) The overall framework of Saliency-Net. (b) The initial saliency prediction module. (c) The semantic segmentation module.

obtain the saliency ground-truth image, which ensures the reliability of the calculated saliency ground truth, as exemplified in Fig. 3.

B. Point Cloud Saliency Prediction

Fig. 4 illustrates the Saliency-Net for point cloud saliency prediction, which comprises two primary components. Fig. 4(a) is the overall network framework, which can be composed of two parts: Fig. 4(b) and (c). Fig. 4(b) is the initial point cloud saliency prediction network. We refer to U-Net [13] using the hourglass encoder and decoder architecture; the encoder is used to obtain the context information of the point cloud picture, while the corresponding decoder can accurately locate the saliency significance area of the point cloud. In addition, we also use residual modules, which have better performance than regular convolutional layers. Each residual module is defined as $2 \times (\text{CONV} + \text{BN} + \text{ReLU})$. Furthermore, the projection image of the LiDAR point cloud is elongated in shape. To avoid losing height information, the downsampling module reduces the dimensions only in the width direction with a stride of 2. The upsampling module uses transposed convolutions with a stride of 2 and combines each upsampled feature map with the corresponding feature map from the encoder to leverage richer feature information at each stage.

Fig. 4(c) represents a semantic segmentation network, whose primary function is to output semantic information of the point cloud. The input of the part (blue horizontal line part in the figure) comes from the feature graph extracted by the network encoder in the first part, while the decoder part adopts the FID and the classification head proposed in FIDNet [14]. In the classification head network, we use dilated convolutions to extract local features and aggregate them by concatenating with the original feature tensor. Finally, a classification layer is used to predict semantic labels.

Finally, the outputs of *b* and *c* are concatenated along the channel dimension, and then a simple downsampling layer

TABLE I
NETWORK PARAMETERS OF SALIENCY-NET

Network	Name	Layer settings	Num.	Output
	Input			$H \times W \times 5$
Initial Saliency Prediction	Conv1	Conv $3 \times 3, 48$	$\times 1$	$H \times W \times 48$
	Conv2	Conv $3 \times 3, 64$	$\times 1$	$\frac{1}{2} H \times \frac{1}{2} W \times 64$
	Conv3	Conv $3 \times 3, 128$	$\times 4$	$\frac{1}{4} H \times \frac{1}{4} W \times 128$
	Conv4	Conv $3 \times 3, 256$	$\times 4$	$\frac{1}{8} H \times \frac{1}{8} W \times 256$
	Conv5	Conv $3 \times 3, 512$	$\times 2$	$\frac{1}{16} H \times \frac{1}{16} W \times 512$
	Deconv6	ConvTranspose $3 \times 3, 256$	$\times 1$	$\frac{1}{8} H \times \frac{1}{8} W \times 256$
	Deconv7	ConvTranspose $3 \times 3, 128$	$\times 1$	$\frac{1}{4} H \times \frac{1}{4} W \times 128$
	Deconv8	ConvTranspose $3 \times 3, 64$	$\times 1$	$\frac{1}{2} H \times \frac{1}{2} W \times 64$
	Deconv9	ConvTranspose $3 \times 3, 128$	$\times 1$	$H \times W \times 1$
Semantic Segmentation	Conv1	Conv $1 \times 1, 48$	$\times 1$	$H \times W \times 48$
	Conv2	Conv $1 \times 1, 48$	$\times 1$	$H \times W \times 48$
	Conv3	Conv $1 \times 1, 48$	$\times 1$	$H \times W \times 48$
	Conv4	Conv $1 \times 1, 48$	$\times 1$	$H \times W \times 48$
	Conv5	Conv $1 \times 1, 48$	$\times 1$	$H \times W \times 48$
	Conv6	Conv $1 \times 1, \text{labels}$	$\times 1$	$H \times W \times \text{labels}$

is applied to obtain the final saliency map for the point cloud. Table I records the specific network parameters of Saliency-Net.

Saliency-Net explicitly integrates multiple point cloud features and is trained under supervision using our designed point cloud dataset with saliency ground truth. Compared to methods using only single information, this approach ensures the completeness of saliency prediction. Even in semantically

sparse scenarios, it effectively predicts saliency by integrating geometric and intensity information, thereby better optimizing SLAM tasks.

In network training, we perform supervised training on two subnetworks using two loss functions. For the point cloud saliency prediction network, its loss function is as follows:

$$L(p, g) = \sum_i |p_i - g_i| \quad (11)$$

$$p_i = \frac{\exp(x_i^p)}{\sum_{i=1}^N \exp(x_i^p)}, \quad g_i = \frac{\exp(x_i^g)}{\sum_{i=1}^N \exp(x_i^g)} \quad (12)$$

where p is the predicted probability distribution of saliency from the network, g is the probability distribution of the ground-truth saliency for the point cloud, and $x = (x_1, \dots, x_i, \dots, x_N)$ is the set of unnormalized saliency values for the predicted saliency map (x^p) and the ground-truth saliency map (x^g).

As for the loss function used in training the semantic segmentation head network, we use the weighted cross-entropy loss function to supervise the network training, as shown in the following formula, M represents the number of categories, the value of y_c is 0 or 1, p_c represents the probability of predicted pixels belong to the category c :

$$L = - \sum_{c=1}^M w_c y_c \log(p_c), w_c = \frac{N - N_c}{N}. \quad (13)$$

When calculating the weight w_c , N represents the total number of pixels, and N_c represents the number of pixels with the true class label being c .

C. Feature Matching and Pose Estimation

The line feature point P_l^c of the current frame is converted to the reference frame as P_l in the reference frame coordinate, the nearest line feature point to P_l in the reference frame is A_l , and the nearest line feature point B_l that locates on the laser scanning line adjacent to the point A_l , the point cloud saliency values of these points are S_{A_l} , S_{B_l} and S_{P_l} . To ensure that the identified points belong to line features within the same local region, we set an error threshold TH_{error} , only when the points satisfied with $|S_{P_l} - S_{A_l}| < \text{TH}_{\text{error}}$ and $|S_{P_l} - S_{B_l}| < \text{TH}_{\text{error}}$ will be chosen

$$e_l = w_l d_l = \frac{(aS_{P_l}^2 + b) \left| \overrightarrow{P_l A_l} \times \overrightarrow{P_l B_l} \right|}{255 \left| \overrightarrow{A_l B_l} \right|}. \quad (14)$$

Different from the tradition that takes the distance d_l as the residual term, it does not distinguish the well-characterized point. We use the $w_l = (aS_{P_l}^2)/255$ as the point cloud saliency weight, where a and b are the constants, giving different point cloud features with different weights.

For the plane feature point P_p^c of the current frame, it is transformed into the reference frame coordinate as P_p , and then a search is performed in the reference frame to find the nearest plane feature point A_p to the P_p . Additionally, the plane feature point B_p on the same LiDAR scanning line as A_p and the plane feature point C_p on the adjacent LiDAR

scanning line are also found, and these three points do not lie on the same line.

$$e_p = w_p d_p = \frac{(cS_{P_p}^2 + d) \left| \overrightarrow{P_p A_p} \cdot (\overrightarrow{A_p B_p} \times \overrightarrow{A_p C_p}) \right|}{255 \left| \overrightarrow{A_p B_p} \times \overrightarrow{A_p C_p} \right|}. \quad (15)$$

Similarly, the point cloud saliency weight is $w_p = (cS_{P_p}^2 + d)/255$, c and d are constants and S_{P_p} is the point cloud saliency value of P_p . So the total loss function is as follows:

$$\text{loss} = \sum_{i=1}^{N_l} e_l^i + \sum_{j=1}^{N_p} e_p^j = \sum_{i=1}^N e(P_i) = \sum_{i=1}^N e(TP_i^c). \quad (16)$$

The key to pose estimation is to iteratively optimize the optimal transformation matrix T , which includes the rotation matrix R and the translation vector t , then calculate the first-order Jacobian matrix J of the residual with respect to the optimization variable

$$J = \frac{\partial(\text{loss})}{\partial T_c^r} = \frac{\partial \left(\sum_{i=1}^N e(T_c^r P_i^c) \right)}{\partial T_c^r} = \sum_{i=1}^N \frac{\partial(e(T_c^r P_i^c))}{\partial T_c^r}. \quad (17)$$

The perturbation ΔT is added to the transformation matrix T by left multiplication, and the derivative is calculated using the Lie algebra approach

$$\frac{\partial(e(TP_i^c))}{\partial \delta \xi} = \frac{\partial(e(P_i))}{\partial(P_i)} \frac{\partial(TP_i^c)}{\partial \delta \xi}. \quad (18)$$

The first part $\partial(e(P_i))/\partial(P_i)$ is the derivative of the residual term $e(P_i)$ on the points P_i , P_i includes the line feature points and plane feature points, the second part $\partial(TP_i^c)/\partial \delta \xi$ is the derivative of point P_i on the Lie algebra $\delta \xi$ of perturbation term ΔT . By calculating the derivative of the residual term with respect to the pose perturbation, the first-order Jacobian of the loss function with respect to the optimization variable can be obtained, which can guide iterative optimization to compute the best pose.

Furthermore, we propose a method capable of effectively suppressing interference from dynamic objects. Specifically, the pose estimation results for frame-to-frame and frame-to-local-map are computed separately. In the frame-to-frame estimation, feature matching is constrained by saliency weights, whereas the frame-to-map estimation is performed without incorporating saliency constraints. If dynamic points exist in the scene, the two estimation approaches will yield significant pose discrepancies. Subsequently, a threshold is applied to evaluate the deviation between these estimates. If the deviation exceeds the threshold, the point is considered potentially dynamic. It is then discarded and excluded from computation, thereby reducing the weight of dynamic points in the pose estimation process.

The pseudocode presented in Algorithm 1 illustrates the entire execution process of our proposed algorithm.

III. BENCHMARK RESULTS

A. Validation of the Effectiveness of the Saliency Dataset

we process the KITTI dataset sequences 00–10 using the same data processing approach and test both the original

Algorithm 1 Feature Matching and Pose Estimation Method Based on Point Cloud Saliency

Input: Point cloud feature P_i in total N , tag features P_i^{label} in total N^{label} , S_i is the point cloud saliency value, $P_i^{label} \in P_i, S_i^{label} \in S_i$ setting the percentage threshold value TH_{per} , and the pose deviation threshold TH_{pose}

Output: The final pose estimation result $pose^f$

- 1: $per = N^{label}/N$;
- 2: **if** $per \leq TH_{per}$ **then**:
- 3: $S_i^{label} = 0$;
- 4: $pose_1 = Esti_{scan}^{scan}(P_i, S_i)$;
- 5: $pose^f = Optim(Esti_{scan}^{map}(P_i, S_i, pose_1), pose_1)$;
- 6: **else**:
- 7: $pose_2 = Esti_{scan}^{scan}(P_i, S_i)$;
- 8: $S_i^{tmp} = S_i^{label}, S_i^{label} = 0$;
- 9: $pose_3 = Esti_{scan}^{map}(P_i, S_i, pose_2)$;
- 10: **if** $(pose_3)^T pose_2 \geq TH_{pose}$ **then**:
- 11: $pose^f = Optim(pose_3, pose_2)$;
- 12: **else**:
- 13: $S_i^{label} = S_i^{tmp}$;
- 14: $pose^f = Optim(Esti_{scan}^{map}(P_i, S_i, pose_3), pose_2)$

TABLE II
ABLATION STUDY ON M2DGR DATASET AND KITTI DATASET

	sequences	w/o Intensity	w/o Semantic	ALL
KITTI	00	0.72	0.64	0.45
	01	0.85	0.91	0.61
	02	0.78	0.63	0.52
	03	0.87	0.75	0.59
	04	0.51	0.58	0.45
	05	0.54	0.46	0.32
	06	0.36	0.41	0.22
	07	0.41	0.51	0.29
	08	1.02	0.87	0.78
	09	0.67	0.72	0.57
	10	0.92	0.72	0.48
M2DGR	01	0.41	0.39	0.28
	02	2.73	3.14	2.61
	03	0.37	0.29	0.13
	04	0.57	0.46	0.39
	05	0.45	0.51	0.41
	06	0.43	0.59	0.33
	07	1.32	1.86	0.31
	08	0.36	0.67	0.28
	09	2.37	2.65	1.51
	10	0.84	1.12	0.24

dataset and the saliency-weighted dataset with state-of-the-art methods. We employ our designed saliency weights to constrain feature matching for optimizing pose estimation. The experimental results, recorded in Table II, demonstrate that the ATE performance of all methods improves with the incorporation of saliency weights, confirming the effectiveness of our saliency dataset.

B. Ablation Study

We conduct ablation experiments on the KITTI dataset and the M2DGR dataset to quantitatively analyze the independent contributions of intensity information and semantic information from point clouds to the performance of Saliency-LOAM. Specifically, for the **w/o Intensity** scheme, we remove the intensity information from the input to the Saliency-Net. As a result, the network relies solely on semantic and geometric information to predict saliency. For the **w/o Semantic** scheme, we disable Fig. 4(c), during training, point cloud image data only passes through Fig. 4(b), which prevents the network from predicting semantic information.

As shown in Table II, regardless of which type of information is removed, the performance falls short of that of the complete Saliency-LOAM. This demonstrates that the intensity information and semantic information of point clouds are complementary and indispensable for identifying more stable feature points within the point clouds, which proves the effectiveness of the point cloud saliency mechanism.

C. State-of-the-Art Comparison

To evaluate the performance disparities between our method and existing state-of-the-art approaches, we conduct tests on the KITTI dataset and the M2DGR dataset. For the KITTI dataset, we choose A-LOAM [15], F-LOAM [16], SUMA++ [17], KISS-ICP [18], MULLS [19], CT-ICP [20], X-ICP [21], KDD-LOAM [22], SG-SLAM [10], and Intensity-SLAM [23]. For the M2DGR dataset, we select A-LOAM [15], LIO-SAM [24], DLO [25], MARS [26], In-LIO [27], Point-LIO [28], X-ICP [21], LOG-LIO [29], SG-SLAM [10], Intensity-SLAM [23], and FAST-LIO2 [30]. Among them, Intensity-SLAM is a method based on intensity information, while SUMA++ and SG-SLAM are methods based on semantic information. We record the results in Tables III and IV respectively.

Furthermore, we compare the computation times of these methods on KITTI Sequence 03, as presented in Table V. If the computational time is less than 100 ms/scan (more than 10 Hz), it indicates the feasibility of achieving real-time localization and mapping. We further decompose the main time-consuming modules of the entire algorithm, including saliency prediction, feature extraction, feature matching and pose optimization, and map update. The results show that saliency prediction and feature matching & pose optimization consume the most time, providing clear directions for developing a more lightweight LiDAR SLAM system in the future.

D. Real-World Experiments

To further validate the practicality of Saliency-LOAM, we conduct experiments in real-world environments. We evaluate the localization performance of various methods on the campus dataset using the absolute trajectory error (ATE) calculated in meters. Table VI presents the performance of existing state-of-the-art methods and our proposed method across all sequences. It can be observed that our method outperforms the others in most sequences, indicating its superior performance in real-world complex environments, which demonstrates the

TABLE III
SLAM PERFORMANCE COMPARISON ON THE KITTI DATASET WITH ATE

Method	00	01	02	03	04	05	06	07	08	09	10	Avg.
A-LOAM	0.78	1.43	0.92	0.86	0.71	0.57	0.65	0.63	1.12	0.77	0.79	0.84
F-LOAM	0.92	2.80	1.56	1.09	1.43	0.79	0.72	0.54	1.54	1.28	1.77	1.31
SUMA++	0.64	1.60	1.00	0.67	0.37	0.4	0.46	0.34	1.1	0.47	0.66	0.7
KISS-ICP	0.54	0.72	0.53	0.63	0.35	0.33	0.26	0.32	0.91	0.52	0.6	0.52
MULLS	0.51	0.62	0.55	0.61	0.35	0.28	0.24	0.29	0.8	0.49	0.61	0.49
CT-ICP	0.51	0.81	0.55	0.78	0.43	0.27	0.28	0.35	0.8	0.47	0.49	0.52
X-ICP	0.51	0.72	0.55	0.71	0.38	0.28	0.31	0.32	0.78	0.48	0.53	0.51
KDD-LOAM	0.53	0.81	0.54	0.67	0.41	0.28	0.29	0.34	0.83	0.50	0.53	0.52
SG-SLAM	0.51	0.84	0.54	0.63	0.44	0.27	0.29	0.31	0.78	0.54	0.49	0.51
Intensity-SLAM	0.68	1.54	0.84	0.81	0.68	0.53	0.59	0.55	1.12	0.65	0.69	0.79
Ours	0.45	0.61	0.52	0.59	0.45	0.32	0.22	0.29	0.78	0.57	0.48	0.48

TABLE IV
SLAM PERFORMANCE COMPARISON ON THE M2DGR DATASET WITH ATE

Method	01	02	03	04	05	06	07	08	09	10	Avg.
A-LOAM	2.35	5.3	0.35	4.83	0.82	0.69	10.93	0.65	11.94	13.06	5.09
LIO-SAM	0.7	3.95	0.14	0.96	0.4	0.42	2.81	2.37	11.51	8.83	3.21
DLO	0.98	3.1	0.14	0.96	0.54	0.35	0.5	0.2	1.93	0.52	0.92
MARS	0.84	5.25	0.15	5.31	14.62	0.89	1.64	0.29	2.77	0.85	3.26
In-LIO	0.37	2.92	0.13	0.49	0.35	0.39	0.35	0.19	1.74	2.24	0.92
Point-LIO	0.59	2.79	0.11	0.41	0.42	0.47	5.78	0.18	1.94	3.43	1.61
X-ICP	0.35	2.66	0.11	0.47	0.35	0.35	0.42	0.21	1.61	0.18	0.67
LOG-LIO	0.28	2.63	0.12	0.49	0.33	0.34	3.17	0.16	3.16	0.39	1.11
SG-SLAM	0.29	2.65	0.12	0.42	0.35	0.35	0.42	0.21	1.57	0.21	0.66
Intensity-SLAM	0.87	2.98	0.21	0.97	0.87	0.43	1.04	0.45	1.65	0.82	1.03
FAST-LIO2	0.37	2.88	0.12	0.49	0.33	0.36	0.34	0.19	1.55	0.18	0.68
Ours	0.28	2.61	0.13	0.39	0.41	0.33	0.31	0.28	1.51	0.24	0.65

TABLE V
COMPARISON RESULTS OF THE COMPUTATIONAL TIME OF ALGORITHMS ON KITTI SEQUENCE 03

Methods	Computation Time (ms/frame)
A-LOAM	84.52
F-LOAM	69.07
KDD-LOAM	58.45
MULLS	63.45
CT-ICP	80.86
Intensity-SLAM	115.32
SG-SLAM	42.15
KISS-ICP	32.03
Ours	92.37
Saliency Prediction	28.76
Feature Extraction	19.13
Feature Matching & Pose Optimization	31.82
Map Update	9.67

effective contribution of intensity and semantic information to the traditional geometric LiDAR SLAM technique.

To validate mapping consistency, we analyze the mapping quality of our method, Fast-LIO2, and X-ICP on Sequence 05 and Sequence 06 of the campus dataset. During the data collection process, we commence from the area marked by the red box, drive around the building complex, and return to the starting point. Therefore, under ideal conditions, the mapped

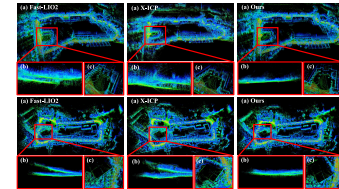


Fig. 5. Comparison of mapping quality on Sequence 05 (top row) and Sequence 06 (bottom row) of the campus dataset. (a) Global map. (b) Degree of overlap between the map at the starting and ending points. (c) Magnified local details.

sections at the starting and ending points should overlap. In Fig. 5(a) represents an example of the global map, (b) illustrates the degree of overlap between the mapped sections at the starting and ending points, and (c) presents an example of local mapping quality. It is observed that, compared to other methods, our method achieves a higher degree of overlap between the starting and ending points, whereas other methods exhibit noticeable deviation. Additionally, the global map constructed by our method demonstrates superior detail.

We also conduct a quantitative analysis comparing our proposed method with other approaches in terms of rotational drift and altitude deviation. The results are recorded in Table VII, and the corresponding data acquisition environment and mapping outcomes are depicted in Fig. 6(a) and (b).

1) *Rotary Drift Test:* As shown in Fig. 6(a), this depicts a round-trip road scenario, starting from landmark P3, reach-

TABLE VI
SLAM PERFORMANCE COMPARISON ON THE CAMPUS DATASET WITH ATE

Methods	01	02	03	04	05	06	07	08	09	10	Avg.
A-LOAM	2.62	1.82	1.59	4.18	1.82	3.79	2.47	1.85	2.22	3.57	2.59
Intensity-SLAM	2.51	1.78	1.64	3.82	1.63	3.25	2.15	1.57	2.01	3.84	2.42
KISS-ICP	2.09	1.54	1.37	3.12	1.43	2.87	1.96	1.32	2.17	3.13	2.10
X-ICP	1.79	1.13	0.73	1.89	1.18	2.24	0.93	1.16	1.76	1.63	1.44
SG-SLAM	1.74	1.35	0.79	1.97	1.22	2.31	0.89	1.21	1.56	1.87	1.49
FAST-LIO2	1.78	1.54	0.81	1.86	1.12	2.17	0.97	1.13	1.43	2.16	1.50
Ours	1.61	1.21	0.65	1.98	0.72	1.64	0.81	1.27	1.14	1.72	1.28

TABLE VII
COMPARISON OF LOCALIZATION PERFORMANCE IN COMPLEX ENVIRONMENTS

Simulate Scene	Metric	Ground Truth	A-LOAM	KDD-LOAM	KISS-ICP	SG-SLAM	X-ICP	Intensity-SLAM	Fast-LIO2	Ours
(a) Rotation Angle	Angle of Experiment [°]	180	167.23	175.84	172.35	163.24	163.45	168.53	173.12	179.18
(b) Altitude Variation	Bias During Experiments [m]	0	0.75	0.69	0.67	0.93	0.71	1.05	0.72	0.63

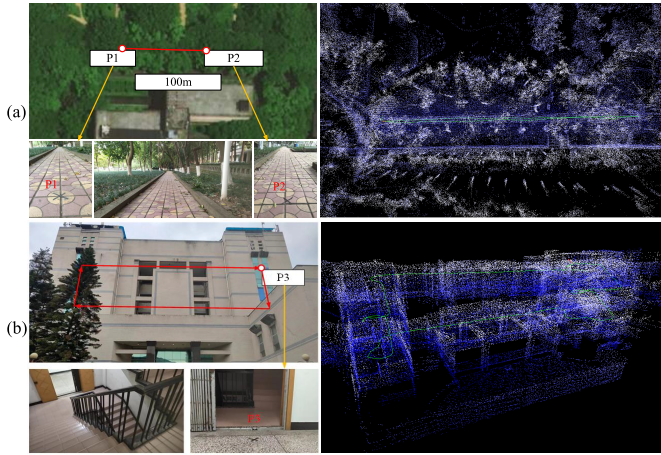


Fig. 6. Results of rotary drift test and height deviation test. (a) Mapping results of large angle rotation scene. (b) Mapping results of height in fast-changing scenes.

ing the U-turn landmark P4, performing a U-turn, and then returning from P4 to P3. At this point, the horizontal angle differs by 180° from the initial angle. This scenario is used to simulate extremely large-angle rotations, which greatly test the rotational robustness of the algorithms. We measure the rotational robustness of each localization algorithm in this scenario by calculating the horizontal angle between the estimated poses at the first and last moments (both at landmark P3) for each algorithm.

2) *Height Deviation Test:* Due to the frequent misplacement of the LiDAR odometry algorithm in the vertical height direction, we conduct a test in a scenario with a two-story height drop to simulate the rapid changes in vehicle height within a short period of time, as shown in Fig. 6(b). Starting from the marked point P5, we descend and ascend once before returning to the origin. Therefore, the true height deviation between the initial and final moments is approximately 0 m. By comparing the height offset errors of the initial and final poses estimated by each method in this scenario.

As indicated in Table VII, our method can fully leverage the surrounding saliency information, resulting in localization outcomes that are closer to the ground truth, thereby achieving more robust and accurate localization.

IV. CONCLUSION

This article proposes a point cloud saliency mechanism that focuses on high-quality point cloud regions, and presents the first LiDAR odometry based on the point cloud saliency. Specifically, we design a deep learning network framework to predict point cloud saliency, considering distance information, local curvature, intensity information, and semantic information. Compared to existing LiDAR odometry, our method emphasizes high-quality point cloud regions and assigns them higher weights during the computation process, reducing the negative impact of noise and unstable points. We validate the effectiveness of the point cloud saliency mechanism through ablation studies and comparative experiments on the KITTI and M2DGR datasets. To demonstrate the practicality of our proposed method, we conduct extensive comparative experiments in real-world environments, and the results show that our LiDAR odometry method outperforms other state-of-the-art methods, significantly improving robustness and accuracy.

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